SYSTEM ARCHITECTURE BLUEPRINT (IEEE 1471 / TOGAF ALIGNED)

Ustad-AI: 3-Tier Edge & Cloud System Architecture

Complete technical specification of Client On-Device Runtime, Local SQLite Engine, Python Gateway, and Cloud Services

OFFLINE-FIRST HYBRID TOPOLOGY

CLIENT TIER (Flutter Native Android)

ON-DEVICE RUNTIME

1. Presentation & UI Layer (Flutter 3.x / Dart)

- **Teacher Mode** Roll Call, AI Oral Exam, STEM Hunt, Paper Grader, SNC Planner, Marks.
- **Admin Mode:** Real-time KPI Dashboard (`fl_chart`), Predictive Risk Briefings, 1-Tap WhatsApp Comms.
- **State Management:** Provider Architecture (Zero UI thread blocking, reactive data binding).

2. Hardware & Operating System Interop Layer

- **Audio Input:** `speech_to_text` (Captures student spoken answers in Urdu/English).
- **Audio Output:** `flutter_tts` (Reads oral exam questions and historical dialogues aloud).
- **Camera Stream:** `camera` & `image_picker` (Captures science specimens for computer vision).
- **OS Intents:** `url_launcher` (Invokes native Android WhatsApp intent with pre-filled URI).

3. Local-First Offline Data Engine (SQLite)

100% OFFLINE

- **Database Helper (sqflite):** Local relational persistence in `ustad_ai.db` .
- **8 Core Tables:** Users, Teachers, Families, Students, Attendance, Marks, Fees, Subjects.
- **Operation:** Zero network dependency for roll call, fee auditing, roster lookup, and grade records.

4. Async Network Client (Dio HTTP Client)

- Handles asynchronous background REST requests to Python backend when internet is active.
- Auto-fallback to local database summaries if offline; exponential backoff and timeout handling.

REST / HTTP

JSON Payload

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Port 8000

BACKEND API TIER

PYTHON FASTAPI

1. REST API Routers

- `/oral_exam` : Evaluates spoken student answer vs rubric.
- `/stem_hunt` : Verifies camera image vs target specimen.
- `/parent_comms` : Generates Urdu WhatsApp text.
- `/roleplay` : Simulates historical figures.
- `/analytics` : Generates predictive risk advice.

2. Service Layer & Safety

- **Safety Guard:** Regex & context sanitation preventing prompt injection.
- **Prompt Engine:** Parameterizes verified student data, school name, and Urdu honorifics ("بیٹی").
- **Vision Processor:** Encodes base64 image buffers for multimodal AI evaluation.

3. Schema Validation

- Pydantic models for strict typed request/response contracts.
- Graceful error handling with structured JSON fallbacks.

HTTPS API

Token Auth

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AI Payloads

EXTERNAL SERVICES

CLOUD & NATIVE

Alibaba DashScope AI

- **Qwen-Max:** State-of-the-art Urdu language comprehension & culturally respectful message synthesis.
- **Qwen3-VL:** ultimodal vision model identifying physical science objects and rejecting false matches.

WhatsApp Messenger

- Launched via native Android Intent: `whatsapp://send?phone=...`
- Pre-fills recipient phone & verified Urdu message with zero copy-pasting.

Deployment Partner

- **Sublime Grammar School:** Gujranwala, Pakistan.
- Dual Parent Access: In-App Mobile Login & 24/7 WhatsApp Agent.
- Validated on real teachers, students, and low-cost Android hardware (2GB-3GB RAM).